Gravitational Sigma Engineering: Feasibility of Local Spacetime Modification via Quantum Information Control

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Within the Σ–τ framework, the gravitational entropy production $\Sigma_{\rm grav} = -\ln(-g_{00})$ is identified as a fundamental field governing the gravitational channel. If $\Sigma_{\rm grav}$ can be modified locally *without* adding mass, one obtains "gravity control" in the strict sense of altering the effective metric. We evaluate four candidate mechanisms quantitatively: (i) Khronon field condensate displacement ($\Delta\Sigma \sim 10^{-97}$), (ii) Casimir vacuum energy ($\Delta\Sigma \sim 10^{-46}$), (iii) superconductor condensation energy ($\Delta\Sigma \sim 10^{-34}$), and (iv) nanoparticle matter-wave interferometry ($\Delta\Sigma \sim 10^{-10}$ for next-generation experiments). All estimates are compared against experimental detection thresholds: torsion-balance sensitivity $\Delta\Sigma \sim 10^{-15}$, atom interferometry $\Delta\Sigma \sim 10^{-20}$, and LIGO-class detectors $\Delta\Sigma \sim 10^{-22}$. We find that approaches (i)–(iii) are quantitatively excluded for laboratory-scale gravity modification, while approach (iv) offers a realistic path to *detecting* $\Sigma_{\rm grav}$ at the quantum scale within the coming decade. We identify the quantum-coherence channel of superconductors as an open theoretical question deserving further investigation.

I. INTRODUCTION

General relativity describes gravity through the spacetime metric $g_{\mu\nu}$, whose curvature encodes the gravitational field [KNOWN]. In the companion papers of this series, we established a quantum-information interpretation: the temporal asymmetry parameter $\tau = 1 - F$, where F is the Petz recovery fidelity, quantifies the irreversibility of the gravitational channel [1]. The Petz recovery bound

$$F \geq e^{-\Sigma/2} \tag{1}$$

relates the recovery fidelity to the entropy production Σ [KNOWN] [2–5]. In Paper 2 [6], we proved the Channel Theorem: for a static gravitational background, the extensive entropy production of the gravitational channel is

$$\Sigma_{\rm grav} = -\ln(-g_{00})\,, \tag{2}$$

which for the Schwarzschild exterior gives $\Sigma_{\rm grav} = -\ln(1 - r_{\rm s}/r)$ [DERIVED].

A key consequence is the Fisher information principle: demanding that $\Sigma_{\rm grav}$ be a harmonic function in vacuum, $\nabla^2\Sigma_{\rm grav} = 0$, yields the exponential metric

$$g_{00} = -e^{-r_{\rm s}/r}\,, \qquad g_{rr} = e^{+r_{\rm s}/r}\,, \tag{3}$$

which is horizon-free and agrees with all current weak-field tests [DERIVED] [6].

This raises a provocative question: if $\Sigma_{\rm grav}$ is a fundamental field rather than merely a derived quantity, can it be modified *locally* without placing mass at that location? Such a modification would constitute "gravity control" in the literal sense—altering the effective metric g_{00} at a chosen point. In this paper, we evaluate this possibility quantitatively, computing $\Delta\Sigma$ achievable by four distinct physical mechanisms and comparing each against experimental detection thresholds. We emphasize that honest assessment requires showing both what works and what does not.

II. DEFINING GRAVITY CONTROL

We define *gravity control* operationally:

Proposition (Gravity control)**.** *A change $\Delta\Sigma \neq 0$ at spatial location $\boldsymbol{r}$, produced by a mechanism that does not introduce additional stress-energy $\Delta T_{\mu\nu}(\boldsymbol{r}) = 0$ at that location.*

From Eq. (2), such a change implies a local metric modification

$$g_{00} \;\to\; g_{00}\, e^{-\Delta\Sigma}\,, \tag{4}$$

and a corresponding change in the local gravitational acceleration

$$\Delta a = \frac{c^2}{2}\,\frac{\partial(\Delta\Sigma)}{\partial r} \tag{5}$$

for a spatially varying $\Delta\Sigma(r)$ [DERIVED].

A. Detection thresholds

The sensitivity of current and planned experiments sets the minimum detectable $\Delta\Sigma$ [KNOWN]:

- **Torsion balance** (Eöt-Wash group [7]): acceleration sensitivity $\delta a \sim 10^{-15}\,{\rm m/s^2}$. Over a baseline $\ell \sim 1\,{\rm cm}$, this corresponds to

$$\Delta\Sigma^{\rm torsion}_{\rm min} \sim \frac{2\,\delta a\,\ell}{c^2} = \frac{2\times10^{-15}\times10^{-2}}{9\times10^{16}} \approx 2\times10^{-34}\,. \tag{6}$$

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However, for a *uniform* $\Delta\Sigma$ over a macroscopic region $L \sim 1\,\mathrm{m}$, the directly measurable quantity is the redshift $\Delta\nu/\nu = \Delta\Sigma/2$, giving a threshold $\Delta\Sigma \sim 10^{-15}$ from optical clock comparisons [8].

- **Atom interferometer** [9]: phase sensitivity $\Delta\phi \sim 10^{-4}$ rad with interrogation time $T \sim 1\,\mathrm{s}$ gives $\delta a \sim 10^{-12}\,\mathrm{m/s^2}$, but over long baselines $L \sim 10\,\mathrm{m}$ the effective $\Delta\Sigma$ threshold reaches $\sim 10^{-20}$.
- **LIGO** [10]: strain sensitivity $h \sim 10^{-22}$ corresponds to transient $\Delta\Sigma \sim 10^{-22}$, but only for oscillatory signals in the detector bandwidth.

III. APPROACH 1: KHRONON FIELD CONDENSATE

In Paper 3 [11], the Khronon action with ghost condensation [KNOWN] [12] is

$$K(Q) = \mu^2(Q-1)^2\,, \tag{7}$$

where $Q = 1/N_\phi$ is the inverse lapse of the Khronon field and $\mu_0 = H_0/c$ is the single free parameter [DERIVED]. The condensate sits at $Q_0 = 1$; a displacement $Q = Q_0 + \delta Q$ produces [DERIVED]

$$\Delta\Sigma = 2\ln(1+\delta Q) \approx 2\,\delta Q \quad (\delta Q \ll 1)\,. \tag{8}$$

The energy density required to sustain this displacement is

$$\rho_\mathrm{K} = \frac{\mu^2 c^2}{8\pi G}\,\delta Q\,(2+\delta Q)\,. \tag{9}$$

With $\mu = H_0/c \approx 7.3\times10^{-27}\,\mathrm{m^{-1}}$ and $G = 6.674\times10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$:

$$\begin{aligned}\mu^2c^2 &= (7.3\times10^{-27})^2\times(3\times10^8)^2\\ &= 5.3\times10^{-53}\times9\times10^{16}\\ &= 4.8\times10^{-36}\ \mathrm{kg\,m^{-1}\,s^{-2}}\,.\end{aligned} \tag{10}$$

For a target $\Delta\Sigma = 10^{-15}$ (optical-clock detectable), we need $\delta Q = 5\times10^{-16}$, giving

$$\rho_\mathrm{K} = \frac{4.8\times10^{-36}\times10^{-15}}{8\pi\times6.674\times10^{-11}} \approx \frac{4.8\times10^{-51}}{1.67\times10^{-9}} \approx 2.9\times10^{-42}\ \frac{\mathrm{kg}}{\mathrm{m^3}}\,. \tag{11}$$

This is not the issue—the energy is negligible. The problem is the *inverse*: the Khronon condensate at cosmological coupling $\mu = H_0/c$ only produces $\Delta\Sigma \sim 10^{-15}$ at the cost of displacing Q on cosmological scales. In the laboratory, with no cosmological horizon to set the boundary condition, δQ relaxes to zero exponentially on the scale $\lambda_K = 1/\mu \sim c/H_0 \sim 10^{26}\,\mathrm{m}$ (the Hubble radius) [DERIVED].

The achievable $\Delta\Sigma$ in a lab of size $L \sim 1\,\mathrm{m}$ is suppressed by $(L/\lambda_K)^2$:

$$\Delta\Sigma_\mathrm{lab} \sim \Delta\Sigma_\mathrm{cosmo}\times\left(\frac{L}{\lambda_K}\right)^2 \sim 1\times\left(\frac{1}{10^{26}}\right)^2 = 10^{-52}\,. \tag{12}$$

Comparing with the Khronon stress-energy contribution over the lab volume $V = L^3$:

$$\Delta\Sigma \sim \frac{8\pi G\,\rho_\mathrm{K}\,L^2}{c^4} = \frac{8\pi\times6.67\times10^{-11}\times\rho_\mathrm{K}\times1}{8.1\times10^{33}}\,. \tag{13}$$

To reach $\Delta\Sigma = 10^{-15}$, one would need $\rho_\mathrm{K} \sim 10^{19}\,\mathrm{kg/m^3}$—nuclear density. But the Khronon condensate at $\mu = H_0/c$ produces $\rho_\mathrm{K} \sim 10^{-26}\,\mathrm{kg/m^3}$ (the critical density), not nuclear density.

Conclusion: Direct Khronon condensation at the cosmological coupling is quantitatively excluded for laboratory gravity modification. Any locally engineered $\Delta\Sigma$ via Khronon displacement would require either (a) a coupling $\mu \gg H_0/c$ (violating the cosmological identification) or (b) boundary conditions mimicking a cosmological horizon at lab scales (physically unrealizable).

IV. APPROACH 2: CASIMIR VACUUM ENERGY

The Casimir effect [KNOWN] [13, 14] produces a negative energy density between parallel conducting plates separated by distance d:

$$\rho_\mathrm{Cas} = -\frac{\pi^2\hbar c}{720\,d^4}\,. \tag{14}$$

For $d = 100\,\mathrm{nm} = 10^{-7}\,\mathrm{m}$:

$$\begin{aligned}\rho_\mathrm{Cas} &= -\frac{\pi^2\times1.055\times10^{-34}\times3\times10^8}{720\times(10^{-7})^4}\\ &= -\frac{9.87\times3.165\times10^{-26}}{720\times10^{-28}}\\ &= -\frac{3.12\times10^{-25}}{7.2\times10^{-26}}\\ &\approx -4.3\ \mathrm{J/m^3} = -4.8\times10^{-17}\ \mathrm{kg/m^3}\,.\end{aligned} \tag{15}$$

The gravitational $\Delta\Sigma$ induced over the plate separation is

$$\Delta\Sigma_\mathrm{Cas} \sim \frac{8\pi G\,|\rho_\mathrm{Cas}|\,d^2}{c^4}\,. \tag{16}$$

Substituting:

$$\begin{aligned}\Delta\Sigma_\mathrm{Cas} &\sim \frac{8\pi\times6.67\times10^{-11}\times4.8\times10^{-17}\times10^{-14}}{8.1\times10^{33}}\\ &\sim \frac{8.0\times10^{-40}}{8.1\times10^{33}}\\ &\sim 10^{-73}\,.\end{aligned} \tag{17}$$

This is extremely small. Even using the Casimir energy over a macroscopic stacking of $N = L/d$ plates over $L = 1\,\mathrm{cm}$ (so $N = 10^5$), the cumulative effect over the full stack is

$$\Delta\Sigma_\mathrm{stack} \sim \frac{8\pi G\,|\rho_\mathrm{Cas}|\,L^2}{c^4} \sim \frac{8\pi\times6.67\times10^{-11}\times4.8\times10^{-17}\times10^{-4}}{8.1\times10^{33}} \tag{18}$$

Conclusion: The Casimir energy density, while much larger than the cosmological Khronon condensate (by ~ 10 orders of magnitude), still produces a gravitational $\Delta\Sigma$ that is ~ 48 orders of magnitude below the optical-clock threshold 10^{-15}. The Casimir route is quantitatively excluded.

V. APPROACH 3: SUPERCONDUCTOR CONDENSATION

A. Energy-based estimate

A type-II superconductor undergoes a phase transition with condensation energy density [KNOWN] [15]

$$u_{\rm cond} = \frac{B_c^2}{2\mu_0}\,, \tag{19}$$

where B_c is the thermodynamic critical field. For $YBa_2Cu_3O_{7-\delta}$ (YBCO) with $B_c \sim 1\,$T:

$$u_{\rm cond} \sim \frac{1}{2 \times 4\pi \times 10^{-7}} \approx 4 \times 10^5 \text{ J/m}^3\,. \tag{20}$$

The corresponding mass-energy density is $\rho_{\rm cond} = u_{\rm cond}/c^2 \approx 4.4 \times 10^{-12}$ kg/m^3.

For a superconducting sample of size $L = 1\,$cm:

$$\begin{aligned}\Delta\Sigma_{\rm SC} &\sim \frac{8\pi G\,\rho_{\rm cond}\,L^2}{c^4} \\ &= \frac{8\pi \times 6.67 \times 10^{-11} \times 4.4 \times 10^{-12} \times 10^{-4}}{8.1 \times 10^{33}} \\ &\sim \frac{7.4 \times 10^{-25}}{8.1 \times 10^{33}} \sim 10^{-58}\,.\end{aligned} \tag{21}$$

More precisely, the *change* in stress-energy upon entering the superconducting state is $\Delta T_{00} = u_{\rm cond}$, and the metric perturbation it sources is of order $h_{00} \sim 8\pi G\,u_{\rm cond}\,L^2/c^4$, yielding the same estimate [DERIVED]. This is ~ 43 orders below threshold.

B. Quantum-coherence hypothesis

A more speculative avenue exists [SPECULATIVE]. In the normal metallic state, conduction electrons undergo frequent scattering events, each producing irreversibility (entropy production). In the Σ–τ framework, each scattering event contributes $\delta\Sigma > 0$. In the superconducting state, Cooper pairs propagate without scattering: $\tau_{\rm electron} = 0$.

If the *gravitational* $\Sigma_{\rm grav}$ couples to the *microscopic* entropy production of matter (beyond the stress-energy tensor alone), then the normal-to-superconductor transition eliminates a source term for $\Sigma_{\rm grav}$.

We estimate the electron scattering contribution to local entropy production. For copper at $T = 300\,$K [KNOWN]:

- Electron density: $n_e \approx 8.5 \times 10^{28}\,\text{m}^{-3}$
- Fermi velocity: $v_{\rm F} \approx 1.6 \times 10^6\,$m/s
- Relaxation time: $\tau_{\rm relax} \approx 2.5 \times 10^{-14}\,$s
- Mean free path: $\ell_{\rm mfp} = v_{\rm F}\,\tau_{\rm relax} \approx 4 \times 10^{-8}\,$m

The entropy produced per electron per scattering is of order

$$\delta\Sigma_{\rm scatter} \sim \frac{G\,m_e^2}{\hbar\,c\,\ell_{\rm mfp}} \tag{22}$$

where $m_e = 9.11 \times 10^{-31}\,$kg. This is the gravitational self-energy of the electron over one mean free path, in natural units:

$$\begin{aligned}\delta\Sigma_{\rm scatter} &\sim \frac{6.67 \times 10^{-11} \times (9.11 \times 10^{-31})^2}{1.055 \times 10^{-34} \times 3 \times 10^8 \times 4 \times 10^{-8}} \\ &= \frac{6.67 \times 10^{-11} \times 8.3 \times 10^{-61}}{1.27 \times 10^{-33}} \\ &= \frac{5.5 \times 10^{-71}}{1.27 \times 10^{-33}} \approx 4.4 \times 10^{-38}\,.\end{aligned} \tag{23}$$

The scattering rate per electron is $1/\tau_{\rm relax} \approx 4 \times 10^{13}\,\text{s}^{-1}$. In a sample volume $V = (1\,\text{cm})^3 = 10^{-6}\,\text{m}^3$, the total entropy production rate is

$$\begin{aligned}\dot{\Sigma}_{\rm total} &= n_e \times V \times \frac{\delta\Sigma_{\rm scatter}}{\tau_{\rm relax}} \\ &= 8.5 \times 10^{28} \times 10^{-6} \times \frac{4.4 \times 10^{-38}}{2.5 \times 10^{-14}} \\ &= 8.5 \times 10^{22} \times 1.76 \times 10^{-24} \\ &\approx 0.15\ \text{s}^{-1}\,.\end{aligned} \tag{24}$$

However, this is the gravitational entropy production *rate* from electron-scattering self-gravity, not the *steady-state* $\Delta\Sigma$ sourced by the scattering. The steady-state requires solving the Poisson-like equation $\nabla^2(\Delta\Sigma) = S$ with appropriate boundary conditions. The source $S \sim G\,n_e\,m_e^2/(\hbar\,c\,\tau_{\rm relax})$ produces a static $\Delta\Sigma$ of order

$$\Delta\Sigma_{\rm coherence} \sim S \times L^2 \sim \frac{G\,n_e\,m_e^2\,L^2}{\hbar\,c\,\tau_{\rm relax}}\,. \tag{25}$$

Substituting:

$$\begin{aligned}\Delta\Sigma_{\rm coherence} &\sim \frac{6.67 \times 10^{-11} \times 8.5 \times 10^{28} \times 8.3 \times 10^{-61} \times 10^{-4}}{1.055 \times 10^{-34} \times 3 \times 10^8 \times 2.5 \times 10^{-14}} \\ &= \frac{6.67 \times 8.5 \times 8.3 \times 10^{-148}}{7.9 \times 10^{-41}} \\ &\sim \frac{4.7 \times 10^{-146}}{7.9 \times 10^{-41}} \sim 6 \times 10^{-106}\,.\end{aligned} \tag{26}$$

This is *far* below any detection threshold, essentially because the electron gravitational self-energy is negligibly small compared to its electromagnetic interactions. The often-cited claims of "gravitational effects in superconductors" [16, 17] would require $\Delta\Sigma \sim 10^{-8}$ to 10^{-5}, which is ~ 100 orders of magnitude above what the electron self-gravity channel can produce.

Conclusion: The energy-based estimate gives $\Delta\Sigma \sim 10^{-58}$; the quantum-coherence route gives $\Delta\Sigma \sim 10^{-106}$. Neither is detectable. However, we stress that Eq. (25) assumes $\Sigma_{\rm grav}$ couples to matter only through gravitational self-energy. If a *non-perturbative* coupling between macroscopic quantum coherence and $\Sigma_{\rm grav}$ exists (for instance, via the topological properties of the Cooper-pair condensate), the estimate could change qualitatively. This remains an open question.

VI. APPROACH 4: NANOPARTICLE INTERFEROMETRY

Recent experiments have achieved matter-wave interference with increasingly massive particles, culminating in the demonstration by Pedalino *et al.* [18] of interference fringes for nanoparticles of mass $m = 170\,\text{kDa} = 2.8\times10^{-22}\,\text{kg}$.

When a massive particle is placed in a spatial superposition

$$|\psi\rangle = \frac{1}{\sqrt{2}}\big(|\boldsymbol{r}_1\rangle + |\boldsymbol{r}_2\rangle\big) \tag{27}$$

with separation $\Delta x = |\boldsymbol{r}_1 - \boldsymbol{r}_2|$, the gravitational self-interaction produces an entropy production

$$\Sigma_{\rm self} = \frac{G\,m^2}{\hbar\,c\,\Delta x}\,. \tag{28}$$

This can be understood as follows [DERIVED]: the gravitational potential energy between the two branches is $E_{\rm grav} = Gm^2/\Delta x$, and the associated $\Sigma_{\rm grav}$ is $E_{\rm grav}/E_{\rm Planck}$ where $E_{\rm Planck} = \hbar c/L^2_{\rm Planck} \times L_{\rm Planck} = \sqrt{\hbar c^5/G}$. More precisely:

$$\Sigma_{\rm self} = \frac{G\,m^2}{\hbar\,c\,\Delta x} = \frac{t_{\rm grav}}{t_{\rm Planck}} \times \frac{L_{\rm Planck}}{\Delta x}\,, \tag{29}$$

where $t_{\rm grav} = Gm/(c^2\Delta x) \times \Delta x/c$ is the gravitational light-crossing time.

A. Current experiments

For the Pedalino *et al.* parameters ($m = 2.8\times10^{-22}$ kg, $\Delta x \sim 10\,\text{nm} = 10^{-8}$ m):

$$\begin{aligned}
\Sigma_{\rm self} &= \frac{6.67\times10^{-11}\times(2.8\times10^{-22})^2}{1.055\times10^{-34}\times3\times10^{8}\times10^{-8}}\\
&= \frac{6.67\times10^{-11}\times7.84\times10^{-44}}{3.165\times10^{-34}}\\
&= \frac{5.23\times10^{-54}}{3.165\times10^{-34}}\\
&\approx 1.7\times10^{-20}\,.
\end{aligned} \tag{30}$$

This is too small to cause observable decoherence on experimental timescales (the decoherence rate scales as $\Gamma \sim \Sigma_{\rm self}\times c/\Delta x$ [19, 20]).

B. Next-generation experiments

Proposals for next-generation experiments target masses $m \sim 10^9$ amu $= 1.7\times10^{-18}$ kg and superposition separations $\Delta x \sim 1\,\mu\text{m} = 10^{-6}$ m [21, 22]:

$$\begin{aligned}
\Sigma^{\rm next}_{\rm self} &= \frac{6.67\times10^{-11}\times(1.7\times10^{-18})^2}{1.055\times10^{-34}\times3\times10^{8}\times10^{-6}}\\
&= \frac{6.67\times10^{-11}\times2.89\times10^{-36}}{3.165\times10^{-32}}\\
&= \frac{1.93\times10^{-46}}{3.165\times10^{-32}}\\
&\approx 6.1\times10^{-15}\,.
\end{aligned} \tag{31}$$

This is tantalizingly close to the optical-clock threshold $\Delta\Sigma \sim 10^{-15}$ and well within the range that could produce *measurable gravitational decoherence.*

The Bose–Marletto–Vedral (BMV) proposal [21, 22] provides a direct experimental test: two particles, each in spatial superposition, interact gravitationally. If they become entangled, this certifies that gravity mediates quantum information—and in our framework, that $\Sigma_{\rm self}$ is a quantum observable. The entanglement phase is

$$\phi_{\rm ent} = \frac{G\,m^2\,t}{\hbar\,\Delta x} = \Sigma_{\rm self}\times\frac{c\,t}{\Delta x}\,. \tag{32}$$

For $\phi_{\rm ent} \sim 1$ (detectable), one needs $t \sim \Delta x/(c\,\Sigma_{\rm self})$. With the next-generation parameters:

$$t \sim \frac{10^{-6}}{3\times10^{8}\times6\times10^{-15}} \sim 0.6\ \text{s}\,, \tag{33}$$

which is within the reach of proposed experiments [21, 23].

C. Interpretation within the Σ framework

If the BMV experiment succeeds, it demonstrates that $\Sigma_{\rm grav}$ at the scale $\Sigma \sim 10^{-14}$ is a genuine quantum-information quantity, not merely a classical entropy. This would validate the central hypothesis of our framework—that $\Sigma_{\rm grav} = -\ln(-g_{00})$ is a fundamental field obeying quantum mechanics—at a scale where its effects are directly measurable.

Conversely, if gravitational decoherence is observed at a rate *exceeding* $\Gamma = Gm^2/(2\hbar\,\Delta x^2)$ (the Diósi–Penrose prediction), this would indicate additional non-unitary contributions to $\Sigma_{\rm grav}$ beyond the self-gravitational channel, potentially pointing to new physics.

VII. FEASIBILITY SUMMARY

Table I collects our estimates.

VIII. DISCUSSION

Our analysis leads to three principal conclusions:

1. Laboratory gravity modification is excluded by many orders of magnitude.—None of the approaches we considered can produce a detectable $\Delta\Sigma$ at a location devoid of additional mass. The fundamental obstacle is the weakness of the gravitational coupling: $G/c^4 \approx 8.3 \times 10^{-45}\,\mathrm{m/J}$, which converts even large energy densities into minuscule metric perturbations. Claims of anomalous gravitational effects in superconductors [16, 17] would require $\Delta\Sigma$ values exceeding our most optimistic estimates by 40–100 orders of magnitude and should be regarded as unsubstantiated.

2. Nanoparticle interferometry is the most promising path to detecting Σ_{grav} at the quantum scale.—Next-generation experiments with $m \sim 10^9$ amu and $\Delta x \sim 1\,\mu$m achieve $\Sigma_{\mathrm{self}} \sim 10^{-14}$, which is *above* the threshold for observable gravitational decoherence and entanglement. The BMV experiment [21, 22] would directly test whether Σ_{grav} is a quantum observable, providing the first empirical anchor for the Σ–τ framework at the quantum-gravitational interface.

3. The superconductor-coherence channel deserves further theoretical investigation.—Our estimate in §V B assumed that Σ_{grav} couples to matter only through the gravitational self-energy of individual electrons. If macroscopic quantum coherence modifies the *boundary conditions* of Σ_{grav} (rather than sourcing it directly), the relevant quantity might be the topological winding number of the order parameter, not the condensation energy. This possibility cannot be excluded by the present analysis and merits a dedicated study within the framework of Paper 6 [24], where the gauge structure of the Σ field is developed.

Finally, we note an important conceptual point. Even though "gravity control" in the engineering sense (generating macroscopic $\Delta\Sigma$ without mass) appears infeasible, the Σ framework provides a new *diagnostic language* for gravitational physics: every gravitational measurement is a measurement of Σ_{grav}, and the Petz bound (1) provides a universal lower limit on the achievable recovery fidelity. This information-theoretic perspective may prove as valuable for gravitational-wave astronomy and precision geodesy as it is for the foundational questions addressed in this paper.

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TABLE I. Feasibility of local Σ modification by four approaches. All numerical values are order-of-magnitude estimates. The detection thresholds assume current best technology; "next-gen" refers to experiments planned for the 2030s.

Approach	$\Delta\Sigma$ achievable	Best threshold	Gap (orders)	Status
Khronon condensate (§III)	10^{-52}	10^{-15}	37	Excluded
Casimir vacuum (§IV)	10^{-63}	10^{-15}	48	Excluded
Superconductor, energy (§V)	10^{-58}	10^{-15}	43	Excluded
Superconductor, coherence (§V B)	10^{-106}	10^{-15}	91	Excluded (but see text)
Nanoparticle, current (§VI)	10^{-20}	10^{-20}	0	Marginal
Nanoparticle, next-gen (§VI)	10^{-14}	10^{-15}	-1 (above)	**Feasible**

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